

LibraryManagement Database

```
create database library;
```

```
Query OK, 1 row affected (0.003 sec)
```

```
MariaDB [(none)]> use library;
```

```
Database changed
```

Authors Table:

```
MariaDB [library]> CREATE TABLE Authors (
```

```
->  author_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
->  author_name VARCHAR(100) NOT NULL
```

```
-> );
```

```
Query OK, 0 rows affected (0.012 sec)
```

```
MariaDB [library]> INSERT INTO Authors (author_name)
```

```
-> VALUES
```

```
-> ('J.K. Rowling'),
```

```
-> ('George Orwell'),
```

```
-> ('R.K. Narayan');
```

```
Query OK, 3 rows affected (0.010 sec)
```

```
Records: 3  Duplicates: 0  Warnings: 0
```

```
MariaDB [library]> select*from Authors;
```

```
+-----+-----+
```

```
| author_id | author_name |
```

```
+-----+-----+
```

```
|      1 | J.K. Rowling |
```

```
|      2 | George Orwell |
```

```
|      3 | R.K. Narayan |
```

3 rows in set (0.001 sec)

Books Table:

```
MariaDB [library]> CREATE TABLE Books (  
  -> book_id INT PRIMARY KEY AUTO_INCREMENT,  
  -> title VARCHAR(200) NOT NULL,  
  -> author_id INT,  
  -> category VARCHAR(100),  
  -> isbn VARCHAR(20) UNIQUE,  
  -> total_copies INT NOT NULL,  
  -> available_copies INT NOT NULL,  
  -> FOREIGN KEY (author_id) REFERENCES Authors(author_id)  
  -> );
```

Query OK, 0 rows affected (0.053 sec)

```
MariaDB [library]> INSERT INTO Books  
  -> (title, author_id, category, isbn, total_copies, available_copies)  
  -> VALUES  
  -> ('Harry Potter and the Sorcerer's Stone', 1, 'Fantasy', '9780439708180', 10, 10),  
  -> ('1984', 2, 'Dystopian', '9780451524935', 5, 5),  
  -> ('Malgudi Days', 3, 'Fiction', '9788185986178', 7, 7);
```

Query OK, 3 rows affected (0.003 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
MariaDB [library]> select*from Books;
```

```
+-----+-----+-----+-----+-----+-----+-----+  
--+
```

book_id	title	author_id	category	isbn	total_copies	available_copies
1	Harry Potter and the Sorcerer's Stone	1	Fantasy	9780439708180	10	10
2	1984	2	Dystopian	9780451524935	5	5
3	Malgudi Days	3	Fiction	9788185986178	7	7

3 rows in set (0.001 sec)

Members Table:

CREATE TABLE Members (

- > member_id INT PRIMARY KEY AUTO_INCREMENT,
- > member_name VARCHAR(100) NOT NULL,
- > email VARCHAR(100) UNIQUE,
- > phone VARCHAR(15),
- > join_date DATE
- >);

Query OK, 0 rows affected (0.044 sec)

MariaDB [library]> INSERT INTO Members

- > (member_name, email, phone, join_date)
- > VALUES
- > ('Rahul Sharma', 'rahul@gmail.com', '9876543210', '2025-01-10'),
- > ('Priya Singh', 'priya@gmail.com', '9876543211', '2025-02-15');

Query OK, 2 rows affected (0.005 sec)

Records: 2 Duplicates: 0 Warnings: 0

MariaDB [library]> select*from Members;

```

+-----+-----+-----+-----+-----+
| member_id | member_name | email      | phone    | join_date |
+-----+-----+-----+-----+-----+
|      1 | Rahul Sharma | rahul@gmail.com | 9876543210 | 2025-01-10 |
|      2 | Priya Singh  | priya@gmail.com | 9876543211 | 2025-02-15 |
+-----+-----+-----+-----+-----+

```

2 rows in set (0.000 sec)

Book Issues Table:

```

MariaDB [library]> CREATE TABLE Book_Issues (
    ->  issue_id INT PRIMARY KEY AUTO_INCREMENT,
    ->  book_id INT,
    ->  member_id INT,
    ->  issue_date DATE NOT NULL,
    ->  due_date DATE NOT NULL,
    ->  FOREIGN KEY (book_id) REFERENCES Books(book_id),
    ->  FOREIGN KEY (member_id) REFERENCES Members(member_id)
    -> );

```

Query OK, 0 rows affected (0.031 sec)

```

MariaDB [library]> INSERT INTO Book_Issues
    -> (book_id, member_id, issue_date, due_date)
    -> VALUES
    -> (1, 1, CURDATE(), DATE_ADD(CURDATE(), INTERVAL 14 DAY));

```

Query OK, 1 row affected (0.005 sec)

UPDATE Books

```

    -> SET available_copies = available_copies - 1
    -> WHERE book_id = 1;

```

Query OK, 1 row affected (0.005 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [library]> select*from Book_Issues;

```
+-----+-----+-----+-----+-----+
| issue_id | book_id | member_id | issue_date | due_date |
+-----+-----+-----+-----+-----+
| 1 | 1 | 1 | 2026-06-20 | 2026-07-04 |
| 2 | 1 | 1 | 2026-06-20 | 2026-07-04 |
+-----+-----+-----+-----+-----+
```

2 rows in set (0.001 sec)

Book Returns Table:

CREATE TABLE Book_Returns (

```
-> return_id INT PRIMARY KEY AUTO_INCREMENT,
-> issue_id INT,
-> return_date DATE NOT NULL,
-> fine_amount DECIMAL(8,2) DEFAULT 0,
-> FOREIGN KEY (issue_id) REFERENCES Book_Issues(issue_id)
-> );
```

Query OK, 0 rows affected (0.066 sec)

MariaDB [library]> INSERT INTO Book_Returns

```
-> (issue_id, return_date, fine_amount)
-> VALUES
-> (1, CURDATE(), 0);
```

Query OK, 1 row affected (0.002 sec)

MariaDB [library]>

MariaDB [library]> UPDATE Books

```
-> SET available_copies = available_copies + 1
-> WHERE book_id = 1;
```

Query OK, 1 row affected (0.003 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [library]> select*from Book_Returns;

return_id	issue_id	return_date	fine_amount
1	1	2026-06-20	0.00

1 row in set (0.001 sec)

View All Books:

select*from Books;

book_id	title	author_id	category	isbn	total_copies	available_copies
1	Harry Potter and the Sorcerer's Stone	1	Fantasy	9780439708180	10	10
2	1984	2	Dystopian	9780451524935	5	5
3	Malgudi Days	3	Fiction	9788185986178	7	7

3 rows in set (0.001 sec)

View issued books with member details:

SELECT

-> bi.issue_id,

```
-> b.title,  
-> m.member_name,  
-> bi.issue_date,  
-> bi.due_date  
-> FROM Book_Issues bi  
-> JOIN Books b ON bi.book_id = b.book_id  
-> JOIN Members m ON bi.member_id = m.member_id;
```

issue_id	title	member_name	issue_date	due_date
1	Harry Potter and the Sorcerer's Stone	Rahul Sharma	2026-06-20	2026-07-04
2	Harry Potter and the Sorcerer's Stone	Rahul Sharma	2026-06-20	2026-07-04

2 rows in set (0.001 sec)